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Question 14 **(Suggested maximum time: 10 minutes)**

The table below shows some information about regular polygons.
These are shapes where all of the angles are the same size.

Number of angles in the polygon	**Part (a)** **Sum** of the angles	**Part (c)** **Size of each** angle
3	180°	60°
4	360°	
5		
6		

(a) The **sum** of the angles increases in a **linear** pattern.

Complete the column in the table above showing the sum of the angles in each of these shapes.

(b) Find a **formula** for the **sum** of the angles in a regular polygon with n angles.
Remember that these values follow a linear pattern.

(c) Complete the column in the table above showing the **size of each** angle in each of these shapes. Remember that, in each polygon, all of the angles are the same size.

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(d) Find a **formula** for the **size of each** angle in a regular polygon with n angles.